

Fig. 72.1 Serial radiographs of a boy with end-stage renal disease with resolution obtained with medical treatment. (A) Age 1 year, severe resorption, disappearance of the radiocarpal edges of the radius and ulna. Delayed bone age 18 months. (B) Age 2 years, resorption in the radius and ulna bones, widening of the intercarpal spaces, healthy radius. (C) Age 3 years, normal bone structure. Bone age 3.10 years. (D) Age 3 years, normal bone structure with appearance of the tibia, fibula, and distal radius. (E) Age 3 years, marked improvement, although some resorption can still be seen.

Fig_72_1

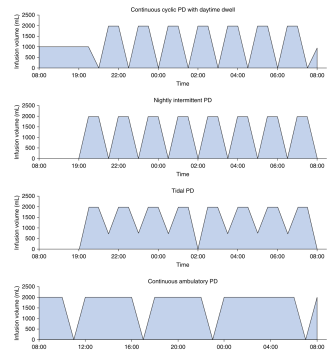


Fig. 72.4 Schematic representation of various peritoneal dialysis regimens based on a standard adult 60 volume of 2000 mL of dialysis fluid. The volume in children is adjusted to the child's body surface area, with an estimated 60 volume being 600 to 1000 mL in older children. The shaded area represents the fill volume and dwell time, and the line represents a 24-hour period. PD, Peritoneal dialysis.

Fig_72_4

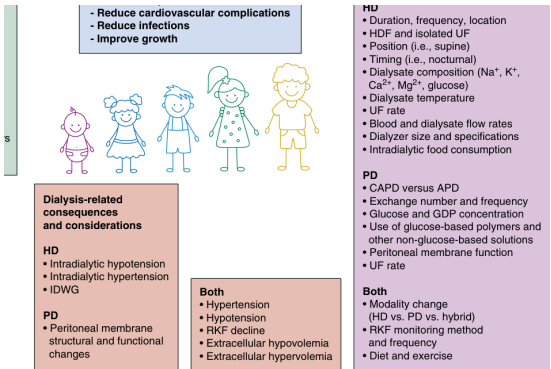


Fig. 72.2 The factors to consider when selecting a dialysis modality.

Fig_72_2

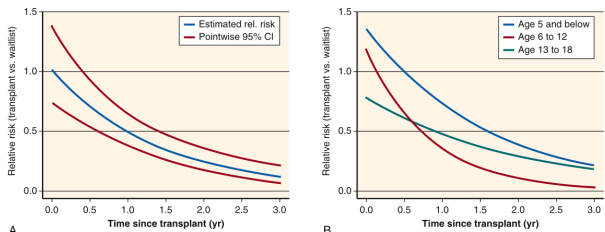


Fig. 72.5 Estimated relative risk of mortality by time since transplantation. (A) The figure is estimated from a model including all ages. (B) Model-based estimates stratified by age. CI, Confidence interval. (Modified from Gillen DL, Stehman-Breen CO, Smith JM, et al. Survival advantage of pediatric recipients of a first kidney transplant among children awaiting kidney transplantation. *Am J Transplant*. 2008;8:2600-2606.)

Fig_72_5

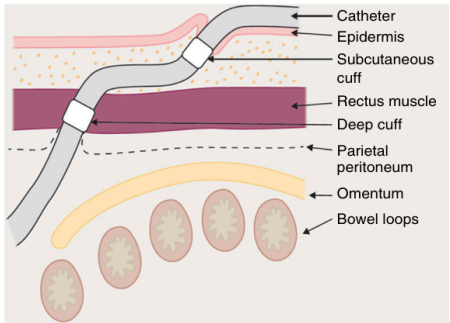


Fig. 72.3 A schematic representation of a peritoneal dialysis catheter and cuff placement in the abdominal wall.

Fig_72_3

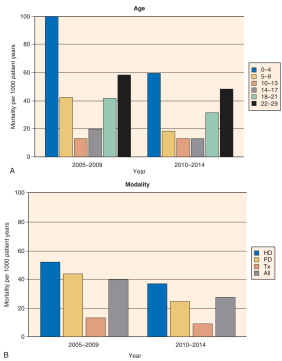


Fig. 72.6 One-year adjusted all-cause mortality rates in incident pediatric patients with end-stage kidney disease for (A) age with comparison to young adults (ages 0-4 years) and (B) modality (ages 0-4 years) and (C) modality (ages 5-9 years) and (D) modality (ages 10-13 years) and (E) modality (ages 14-17 years) and (F) modality (ages 18-21 years) and (G) modality (ages 22-25 years). (Data from 2017 USRDS Annual Data Report. Chapter 7. USRDS among children, adolescents and young adults. doi:10.26907/2374-3658.0808-0918)

Fig_72_6

Table 72.1 Criteria for Defining Chronic Kidney Disease (CKD)	
Criteria for CKD (Either of the Following Must Be Present for >3 Months)	
Markers of kidney damage	- Albuminuria (>30 mg/24 h) or albumin:creatinine ratio >30 mg/g (3 mg/mmol) - Ultrasound evidence of structural abnormalities - Disease-specific mutations detected by genetic analysis - Electrolyte abnormalities and acidosis/alkalosis due to tubular disorders - Abnormalities detected on histology - History of kidney transplantation
Decreased GFR	GFR <60 mL/min/1.73 m ² (GFR G3a–G5)

Table_72_1

Table 72.2 Clinical Consequences and Key Aspects of the Management of Chronic Kidney Disease (CKD) in Children	
Hypertension	- Individual BP targets are based on age, height, and comorbidities - Low salt diet and normalize weight children with salt-loading disease such as CKD - Low salt diet, especially CKD, in the early stages of CKD - Regular physical exercise - Minimize screen time
Chronic Hyperkalemia	- Delay potassium restriction - Low dietary and potassium-binding resins should be administered or dose adjustments of RAAS inhibitors and aldosterone antagonists must be considered
Mineral and bone disease	- Delay restriction of phosphate from early stages of CKD - Vitamin D supplementation with monitoring of 25-hydroxyvitamin D levels - Phosphate binders. In CKD G3a–G5 (including those on dialysis), lower elevated phosphate levels toward the normal range but avoid hypophosphatemia - For patients on dialysis, PTH lowering therapy, vitamin D analogs and calcimimetics are recommended
Proteinuria	- RAAS inhibition with angiotensin-converting enzyme inhibitors (ACEIs) or angiotensin receptor blockers (ARBs) - Lower intraglomerular pressures including sodium-glucose cotransporter 2 inhibitors (SGLT2) are under investigation in children
Metabolic Acidosis	- Diet bicarbonate - Maintain bicarbonate levels within normal range (>21 mmol/L)
Anemia	- It should be maintained within normal range for age - Erythropoiesis-stimulating agents (ESAs) - Iron supplementation, oral or intravenous - ESAs and transfusion - Avoid blood transfusions, especially in potential transplant recipients to avoid sensitization

Table_72_2

Table 72.3 Risk Factors for Growth Failure in Children with Chronic Kidney Disease (CKD)	
Risk Factor	Pathophysiology
Degree of CKD	Standardized height worsens with the progression of CKD. A strong correlation is present between eGFR and height (Z-score) <-1.8, -1.5 and -0.9 for GFR <15, 15–25, 25–30, and >30 mL/min/1.73 m ² , respectively
Malnutrition	Due to a combination of altered taste sensation, altered clearance of cytokines that affect appetite and satiety, electrolytes, and fluid imbalance
CKD-MBD	Results from high FGF23 levels, hypophosphatemia, 25-hydroxyvitamin D deficiency, reduced production of 1,25-dihydroxyvitamin D, hypocalcemia, hyperparathyroidism, malnutrition and drug toxicity (corticosteroids, calcitonin inhibitors)
Metabolic acidosis	Due to the decrease in total renal ammonium excretion when the GFR falls to <40–50 mL/min/1.73 m ² (CKD G3)
Anemia	Low erythropoietin production, reduced lifespan of red cells, and impaired intestinal iron absorption
Endocrine dysfunction	Resistance to growth hormone (GH), modifications of the GH/pituitary-like growth factor 1 (IGF-1) axis
Underlying syndrome	Genetic disorders such as trisomy 21, Turner syndrome, and multiple inherited disorders that affect kidneys
Treatments	Example: corticosteroids

Table_72_3

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Modality	Advantages	Disadvantages
Peritoneal dialysis	Increased patient freedom Longer preservation of residual kidney function Absence of vascular access Improved school attendance Decreased dietary restrictions Technically more feasible in smaller children Possible for patients who live far away from HD center Lower cost	Dependent on parental/caregiver participation Caregiver burden Medicalization of the home Infection Risk of peritoneal membrane failure if on PD for >5 years
In-center hemodialysis	Long-term technique survival Suitable if patients/parents/caregivers unable to perform dialysis at home Decreased treatment time	Limited by availability of vascular access Availability dependent on local resources Increased cost when compared with other modalities
Home hemodialysis	Increased flexibility with dialysis times Improved school attendance Reduced intradialytic symptoms and hypotensive episodes Improved cardiovascular outcomes compared with ICHD More cost-effective compared with ICHD Improved appetite, growth and energy	Dependent on parental/caregiver participation Caregiver burden More rapid loss of residual renal function Increased fistula complications Medicalization of the home Increased home utility bills (water, electricity)

HD, Hemodialysis; ICHD, in-center hemodialysis; PD, peritoneal dialysis.

Table_72_4

Patient Weight	Temporary CVCs	Tunneled CVCs
<10 kg	6.5 Fr 10 cm	8 Fr 18 cm (double-lumen)
10-20 kg	8.0 Fr 10 cm	10 Fr 15 cm (split); 8 Fr 18 cm (double-lumen)
10-20 kg	8.0 Fr 12.5 cm	10 Fr 18 cm (split); 14.5 Fr 13 cm (double-lumen)
20-30 kg	11 Fr 12.5 cm	14 Fr 24 cm; 14.5 Fr 19 cm (double-lumen)
>30 kg	11 Fr 15 cm	14 Fr 24 cm; 14.5 Fr 23 cm (double-lumen)

Table_72_5

Dialyzer Characteristic	Range of Options	Considerations
Surface area	0.2 m ² -2.5 m ²	Larger surface gives greater capacity for clearance and ultrafiltration but requires high blood flow to achieve these advantages
Priming volume	18 mL-140 mL	Consider along with tubing set volume to determine maximum safe extracorporeal volume, especially for smaller patients
Membrane material	Cellulosic, modified cellulose, artificial	Greater biocompatibility with modified cellulose and further with artificial (plastic) membranes
Sterilization	Ethylene oxide (ETO), γ irradiation, steam, electron beam	Allergic reactions associated with ETO; fewer reactions with γ and steam
Flux	Low flux vs. high flux	Higher-flux dialyzers have higher permeability (e.g., larger pores), permitting higher rates of middle molecule clearance and higher UF rate at a given prescription
Clearance (typically described as K _D for urea [mL/min] at blood flow of 200 mL/min)	K _D A Urea 170 mL/min-1700 mL/min	Varies with surface area, flux, membrane material
Ultrafiltration capability (K _{UF} mL/h/mm Hg)	K _{UF} 7 mL/h/mm Hg-111 mL/h/mm Hg	Varies with surface area, flux, membrane material

Table_72_6

Name	Company	Surface Area (m²)	Volume (mL)
Dialyzers			
FX Paed	Fresenius	0.2	18
Sureflux 03L	Nipro	0.3	25
Sureflux 05E	Nipro	0.5	35
FX40	Fresenius	0.6	32
Sureflux 07E	Nipro	0.7	45
Sureflux 09E	Nipro	0.9	55
FX50	Fresenius	1.0	59
Sureflux 11E	Nipro	1.1	65
Sureflux 13E	Nipro	1.3	78
FX60	Fresenius	1.4	74
FX80	Fresenius	1.8	95
Tubings			
Neonatal	Gambro		33
Pediatric	Gambro		85
Adult	Gambro		132
Paed AV set ONLINEplusBVM	Fresenius		111
Paed SN set ONLINEplusBVM	Fresenius		142
Adult HDF	Gambro		151
Adult AV set + BVM	Fresenius		192

*Several different makes and sizes of dialyzers are available globally; this is an example only.
 AV, Arteriovenous; BVM, blood volume monitoring; HDF, hemodiafiltration; SN, single needle.

Table_72_7

[illegible]

Table_72_8

Absolute Contraindications	Relative Contraindications
Omphalocele or gastroschisis	Impending abdominal surgery
Bladder extrophy	Presence of ileostomies and colostomies
Diaphragmatic hernia	Significant organomegaly such as with polycystic kidney disease
Peritoneal membrane failure	Lack of suitable caregiver
Obliterated peritoneal cavity	Inadequate living situation for home dialysis

Table_72_9

Modality	Technique	Advantages	Disadvantages	Indication
ART	Compiling machine code	- Fastest execution - Reduced control overhead - Reduced overhead of context switching	- Overhead of context switching - Overhead of managing multiple threads	Real-time applications
		- Flexibility (any program) - Flexibility of scheduling - No interference or contention of resources (program control)		
COFO	34 bit dispatch	- Preemptive additional context switching - Reduced overhead of context switching	- Disruption of the program - Changes in the program	Highly interactive real-time systems
	Dispatch delay with ~ 80% RTN	Reduced overhead		
IFO	- Dispatch delay based on deadline - Dispatch delay based on priority 1 to 16 bit dispatch	Improved accuracy	- Increased cost of context switching - Complex scheduling mechanism - Complex scheduling mechanism	High-resolution real-time systems
	Priority control based on 3-bit priority	Reduced cost of context switching	- Reduced overhead of context switching - Reduced overhead of context switching	Real-time systems
	Priority control based on 3-bit priority	Reduced overhead of context switching	- Reduced overhead of context switching - Reduced overhead of context switching	Real-time systems
Total	- Natural fit of program into a computer - Program control overhead - Fast generally priority, not a dispatch delay	- Reduced overhead of context switching - Reduced overhead of context switching - Reduced overhead of context switching	- Reduced RTN - Reduced overhead of context switching - Reduced overhead of context switching	- Real-time systems - Real-time systems - Real-time systems
COFO	34 bit dispatch - Long word control (32 bit)	- Preemptive additional context switching - Reduced overhead of context switching	- Disruption of the program - Changes in the program	- Highly interactive real-time systems - Highly interactive real-time systems

Table_72_10

	Age at Transplantation				
	0-1 Years (Percent)	2-5 Years (Percent)	6-12 Years (Percent)	13-17 Years (Percent)	>18 Years (Percent)
Gender					
Male	70.0	65.2	58.0	56.0	54.3
Female	30.1	34.8	42.0	44.0	45.7
Population					
Non-Latino	73.1	62.3	60.0	55.7	50.9
European	7.9	14.6	14.9	19.7	24.5
African American	12.7	16.1	17.8	17.9	16.8
Hispanic (or Latino)	6.3	7.0	7.2	6.7	7.7
Other					
Primary Diagnosis					
Renal dysplasias	27.5	23.3	16.4	11.4	9.8
Obstructive uropathy	19.6	20.2	15.6	13.2	9.9
Diabetes	52.4	48.7	55.6	62.3	63.9
FSGS	0.6	7.7	12.4	13.1	16.4

Since 1987, data have been collected on a total of 12,920 renal transplants in 11,870 pediatric patients. This report represents 1317 new transplants and 1238 new patients entered into the transplant registry since the 2010 Annual Report published in 2014. Of the 12,920 transplants, 10,032 were primary transplants and 2888 were repeat transplants. Adapted from the 2018 Transplant Report of the North American Pediatric Renal Trials and Collaborative Studies (NAPRTCS).¹⁸⁸

Table_72_11

Table 72.12 Evaluation of Pediatric Candidates for Renal Transplantation	
Medical history	End-stage renal disease etiology, family history of renal or other disease, biopsy results, previous transplant history, dialysis access and prescription, medication list, compliance with medical regimen, urologic interventions, urine output, other comorbidities, allergies, previous procedures, transfusions, growth charts
Physical examination	Weight, height, body mass index and percentiles, blood pressure and pulse, general well-being, complete physical examination
Specialist assessments	Pediatric nephrologist, transplant surgeon, urologist, anesthesiologist, transplant coordinator, nurse, social worker, dietitian, pediatric dentist. As indicated: psychologist, cardiologist, hematologist, pulmonologist
Laboratory tests	Blood type, complete blood count, electrolytes, blood urea nitrogen, creatinine, calcium, phosphorus, liver enzymes, protein, albumin, lipids, iron, parathyroid hormone, thyroid, fasting glucose and HbA1c, prothrombin time/partial thromboplastin time, urinalysis, 24-h urine collection for creatinine clearance and proteinuria
Serology	Epstein-Barr virus immunoglobulin [IgG Epstein-Barr nuclear antigen and IgM], cytomegalovirus [IgG and IgM], hepatitis B virus (surface Ag and Hb), hepatitis C virus and human immunodeficiency virus, varicella, purified protein derivative test
Compatibility testing	Human leukocyte antigen typing—class I (A,B), class II (DR, DQ), panel-reactive antibodies, donor-specific antibodies, crossmatch with donor (T- and B-cell crossmatch, by complement-dependent cytotoxicity with antihuman globulin or flow cytometry)
Imaging	Chest radiograph, abdominal ultrasound including the kidneys, bladder, and postvoiding volume Cardiac echo and electrocardiogram. Doppler study of abdominal and pelvic arteries and veins Voiding cystourethrogram, urodynamic testing, or other if indicated Diphtheria, tetanus, pertussis (DTaP), hemophilus influenza type B (Hib), inactivated polio vaccine (IPV), hepatitis A virus (HAV), hepatitis B virus (HBV), influenza, measles, mumps, and rubella (MMR), pneumococcus conjugate and polysaccharide, influenza
Vaccinations	Selected patients: meningococcus, human papillomavirus
Social	Evaluation of the family, support systems, financial issues, school

Table_72_12